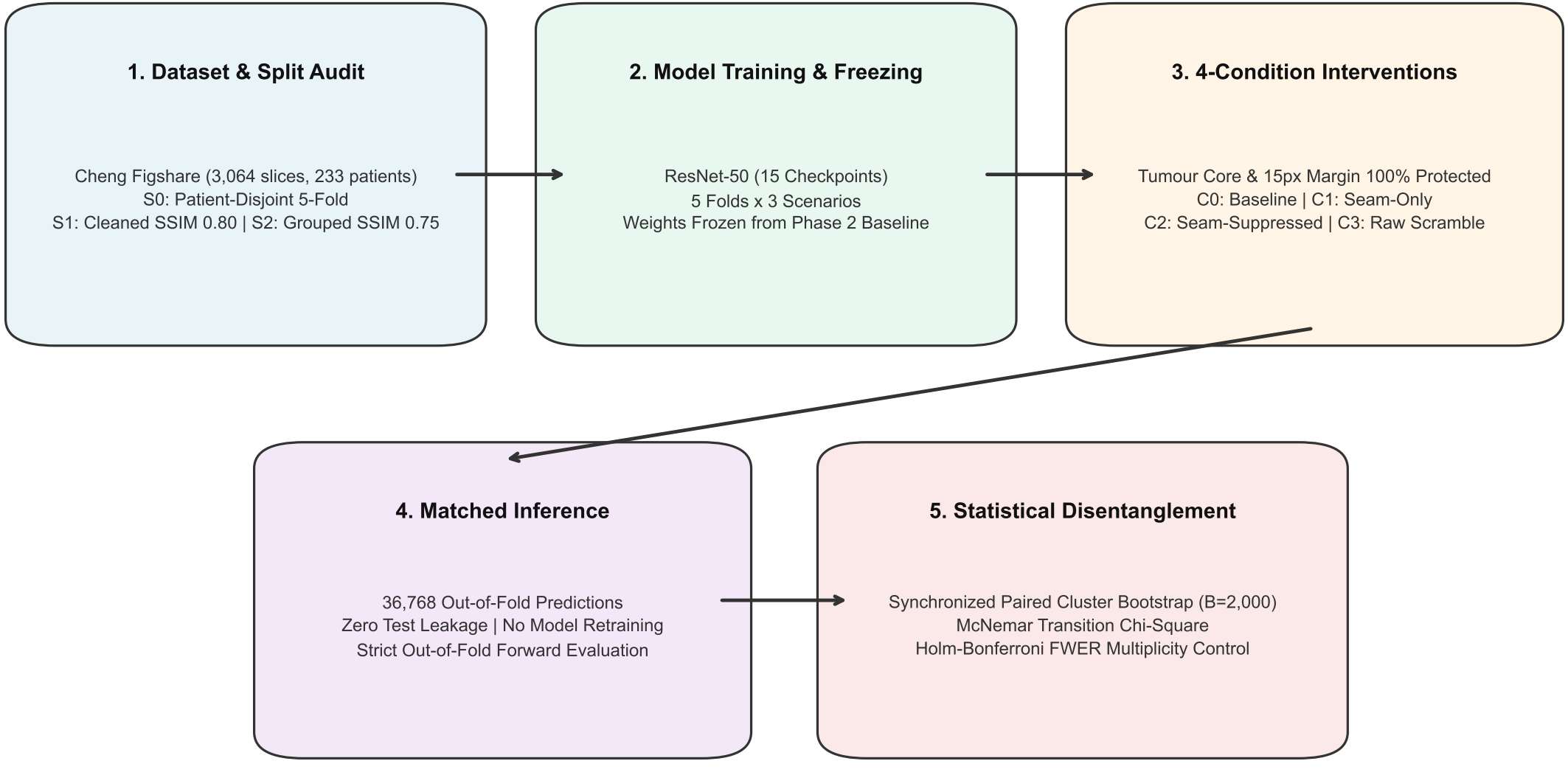
1. Dataset & Split Audit
Cheng Figshare (3,064 slices, 233 patients)
S0: Patient-Disjoint 5-Fold
S1: Cleaned SSIM 0.80 | S2: Grouped SSIM 0.75
2. Model Training & Freezing
ResNet-50 (15 Checkpoints)
5 Folds x 3 Scenarios
Weights Frozen from Phase 2 Baseline
3. 4-Condition Interventions
Tumour Core & 15px Margin 100% Protected
C0: Baseline | C1: Seam-Only
C2: Seam-Suppressed | C3: Raw Scramble
4. Matched Inference
36,768 Out-of-Fold Predictions
Zero Test Leakage | No Model Retraining
Strict Out-of-Fold Forward Evaluation
5. Statistical Disentanglement
Synchronized Paired Cluster Bootstrap (B=2,000)
McNemar Transition Chi-Square
Holm-Bonferroni FWER Multiplicity Control